MedTray: Smart Medication & Medicine Dispenser

An Integrated UI/UX Case Study for Elderly Medication Management in Home Environments

Introduction

Despite the growth of digital health technologies, medication management for elderly patients at home is still mostly manual and prone to errors. Many older adults forget doses, misread labels, or take medicines incorrectly, while caregivers struggle to monitor adherence and pharmacists have limited control once prescriptions are dispensed.
Existing solutions like pillboxes or phone reminders solve only part of the problem and do not connect patients, caregivers, and pharmacists in one system. This report presents the UX process behind MedTray, a smart medication dispenser designed to reduce errors, lower memory load, and make elderly medication management at home safer and easier.

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Section 01 — Problem Definition

Medication management is essential for many elderly individuals living independently with chronic conditions. However, age-related challenges such as memory decline, reduced vision, and reduced motor ability make managing multiple medicines difficult and risky.
MedTray is a Smart Medication Management System designed to support elderly users at home by improving safety, accessibility, and medication adherence.

1.1 What Problem Are We Solving?

Elderly individuals often take multiple medications at different times with specific instructions. Managing these schedules requires memory and attention, which may decline with age.
Common problems include:

- Forgetting doses
- Taking the wrong medicine
- Incorrect dosages
- Double-dosing
- Misunderstanding pharmacist instructions

These errors can worsen health conditions and increase hospital visits, especially when patients live alone.
Current systems rely on memory, handwritten notes, or basic tools like pillboxes and alarms. These solutions do not fully prevent mistakes because users must still identify and manage medicines themselves.

Core Problem:
How can we design a safe and accessible system that reduces medication errors for elderly users at home?
MedTray addresses this by:

- Digitally sharing prescriptions with caregivers
- Allowing remote monitoring
- Providing reminder alerts
- Automatically dispensing correct doses
- Reducing cognitive load

The goal is to improve safety and independence.

1.2 Why Does It Matter?

Medication errors among elderly users directly affect health and quality of life. Many older adults live independently, while caregivers may not be physically present daily.
In real home settings:

- Patients may forget if they took medicine
- Caregivers cannot continuously monitor

- Instructions may be forgotten
- Similar packaging causes confusion

Even small mistakes can lead to serious consequences. Caregivers also experience stress due to lack of visibility.

This issue matters because:

- Medication adherence is critical for chronic diseases
- Preventable errors increase healthcare costs
- Elderly individuals deserve safe independent living
- Caregivers need reassurance

MedTray connects pharmacists, caregivers, and patients through a simple smart system that improves communication and reduces risk.

1.3 Scope (Included / Not Included)

This project focuses on medication management for elderly users in home environments.

Included in Scope

- Digital sharing of prescription details
- Caregiver mobile app for schedule management and monitoring
- Smart medication dispenser with reminders
- Automatic dose dispensing
- Accessibility-focused interface (large text, simple interaction)
- Basic adherence tracking

Not Included in Scope

- Hospital systems
- Emergency services integration
- AI diagnosis or prescription decisions
- Full electronic health records
- Insurance or payment systems

MedTray supports home-based medication adherence but does not replace healthcare professionals.

Section 02 — User Understanding

2.1 Target Users and Context of Use

MedTray is designed for home-based medication management and involves three main user groups: elderly patients, caregivers, and pharmacists. Elderly patients and caregivers use the system regularly, while pharmacists support it during the prescription stage.

Elderly Patients (Primary Users)

Elderly patients use the smart medication device at home. Many live independently and manage multiple daily medications.
Their context includes:

- Memory decline and difficulty remembering schedules
- Reduced vision or hearing
- Limited familiarity with technology
- Fear of making medication mistakes

The device must therefore be simple, clear, and easy to operate.

Caregivers (Secondary Users)

Caregivers may be family members or professionals who support the patient's routine. They are often busy and cannot always monitor the patient physically.
Their context includes:

- Managing schedules remotely
- Receiving alerts for missed doses
- Needing reassurance about adherence
- Reducing stress related to medication errors

The mobile app provides monitoring and notifications.

Pharmacists (Supporting Users)

Pharmacists interact with the system when dispensing medication. Their role is to provide accurate prescription details digitally.
Their context includes:

- Working in busy pharmacy settings
- Requiring a quick and reliable input process
- Reducing miscommunication and errors

Understanding these users ensures MedTray supports safe, coordinated, and simple medication management at home.

2.2 Key Pain Points and User Needs

Based on elderly healthcare research and common medication adherence challenges, several key pain points and user needs were identified.

1. Memory Limitations & Cognitive Overload

Elderly users often forget whether they took a dose, the correct time, or which medicine to take. Managing multiple medications with different schedules increases mental burden and confusion.

Need:

- Automated reminders
- Clear, simple instructions
- Reduced cognitive load

2. Medication Errors & Fear of Mistakes

Common errors include taking the wrong pill, incorrect dosage, wrong timing, or double dosing. Many elderly users also fear making such mistakes, which increases anxiety and reduces independence.

Need:

- Automatic and controlled dispensing
- Time-locked access to prevent double dosing
- Clear confirmation of correct intake

3. Accessibility Challenges

Many elderly individuals experience reduced vision, hearing difficulties, and limited motor control. Complex smartphone apps with multiple steps often lead to frustration and avoidance.

Need:

- Large text and clear displays
- Audio alerts
- Simple one-button interaction
- Minimal and intuitive interface design

4. Caregiver Stress & Lack of Visibility

Caregivers cannot always supervise patients physically and often worry about missed or incorrect doses.

Need:

- Real-time notifications
- Adherence tracking
- Easy remote schedule management

Section 03 - Personas / User Types

Elderly Patient Persona

Mary Perera

Age: 72 | **Gender:** Female

Occupation: Retired Teacher

Mary is a 72-year-old retiree who wants to take her medicines correctly and independently. She sometimes forgets timings and finds small labels difficult to read, so she prefers simple, clear systems that guide her through each dose safely

Goals

- Take her medicines correctly and on time without relying on others.
- Maintain independence and manage her daily health confidently.
- Avoid medication mistakes such as taking the wrong pill.
- Feel safe and reassured while managing her medicines at home.

Frustrations

- Forgetting medicine times occasionally.
- Difficulty reading small labels on bottles.
- Confusion when multiple pills look similar.
- Anxiety about taking the wrong dose.

Context

- Lives at home, sometimes with family support.
- Takes multiple medicines daily.
- Limited experience with technology.
- Prefers simple, clear, and reliable devices.

Quotes

- "I don't want to depend on others every time I take medicine."
- "These labels are too small for me to read properly."
- "I hope I remember to take the right pill."
- "Simple reminders make me feel confident."

Caregiver Persona

Shenali Fernando

Age: 34 | **Gender:** Female

Occupation: Caregiver

Shenali is a 34-year-old working professional who helps manage her parent's medicines while balancing a full-time job. She prefers a system that monitors adherence, sends alerts, and reassures her about patient safety.

Goals

- Ensure the patient takes medicines on time every day.
- Monitor adherence remotely without constant supervision.
- Reduce stress and worry about missed doses.
- Manage caregiving alongside her work efficiently.

Frustrations

- Cannot always be present to remind the patient.
- Unsure whether doses are taken correctly.
- Manually tracking medicines is time-consuming.
- Anxiety about patient safety when away.

Context

- Balances work and caregiving duties.
- Uses smartphone apps regularly.
- Organizes medicines and reminders.
- Communicates with doctors or pharmacists occasionally.

Quotes

- "I just want to know she took her medicine."
- "I worry when I'm not around."
- "It's hard to manage everything myself."
- "I need a reliable system to help me."

Section 04 - Task Flow / User Journey

Task flow

This flow covers the complete lifecycle of managing the medication, highlighting the divergence between manual entry and pharmacist integration.

1. Device Pairing (Caregiver)
    - Open App -> Navigate to Connect Your Device screen.
    - Plug in the physical dispenser.
    - Tap Search for Device to establish an internet/Bluetooth connection.
    - Connection successful -> Reroute to Home dashboard.

2. Schedule Creation (Two Paths)
    - Path A: Pharmacist Integration (Optional)
        - Pharmacist inputs prescription into their system.
        - System sends schedule via email to Caregiver.
        - Caregiver opens App -> Navigates to Care tab.
        - Caregiver reviews "New Rx" or "Adjusted" cards.
        - Caregiver taps Accept & Update.
    - Path B: Manual Caregiver Entry
        - Caregiver opens App -> Navigates to Routine tab.
        - Caregiver taps the + Floating Action Button.
        - Caregiver searches and selects a pill on the Add Medication screen.
        - Caregiver configures the dose and reminder time (e.g., 2 pills at 08:00 AM).

3. Hardware Assignment (Caregiver)
    - System prompts Caregiver to navigate to The Tray screen.
    - Caregiver views the circular 12-slot visual map.
    - Caregiver taps an empty slot (e.g., Slot 6).
    - Caregiver taps + Add Medication to link the digital schedule to that physical slot.
    - Caregiver physically places the correct pills into the matching slot in the machine.

4. Daily Dispensing (Patient)
    - Scheduled time arrives (e.g., 08:00 AM).
    - Dispenser alarm sounds and the central green LED button glows.
    - Patient approaches the machine and pushes the green button.
    - Machine dispenses medicine into the tray and logs the status as "Taken".
    - Caregiver’s Home timeline updates with a green checkmark.

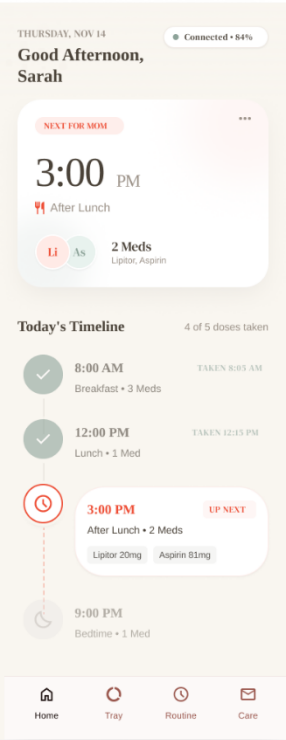
5. Exception Handling (Caregiver)
    - Patient fails to press the button within the allotted timeframe.
    - App triggers a push notification to the Caregiver.
    - Caregiver opens App -> Views Alert Modal (Mom missed the 9:00 AM dose).
    - Caregiver selects an action: Call Mom or Log as Taken.

User Journey

Phase	Actor	Goal	Touchpoint	Emotional State / UX Focus
Prescribing	Pharmacist	Securely and accurately transmit medication instructions to the caregiver.	Pharmacy System & Email	Focus: Efficiency and accuracy. The UI must ensure no transcription errors occur.
Onboarding	Caregiver	Connect the hardware to the phone without technical frustration.	App: Connection Screen	State: Anxious but hopeful. Focus: Clear, step-by-step instructions (e.g., "pulse orange").
Scheduling	Caregiver	Consolidate manual entries and pharmacist updates into one cohesive routine.	App: Care & Routine Tabs	State: Focused. Focus: Clarity. Differentiating between "Action Required" prescriptions and active routines.
Loading	Caregiver	Physically fill the machine without placing the wrong pill in the wrong slot.	App: Tray Screen & Physical Machine	Focus: Confidence. The visual mapping of the circular tray UI directly mirroring the physical device is critical here.
Dispensing	Patient	Receive the right medication at the right time without needing to remember schedules or read small labels.	Physical Machine (LED Button & Alarm)	State: At ease. Focus: Accessibility. Zero friction; a single illuminated button push handles the entire physical interaction.
Monitoring	Caregiver	Ensure the patient is safe and adherence is high, intervening only when necessary.	App: Home Timeline & Push Notifications	State: Reassured. Focus: Peace of mind. Clear visual indicators (green checkmarks vs. red alert modals) for quick scanning.

Section 05 - Design Solution

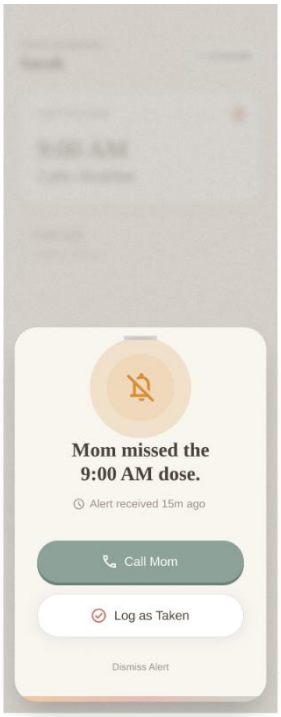
Wireframes



Wireframe 1: Home Page

Design Decision: A prominent "Next Up" Hero Card at the top, supported by a vertical timeline showing completed vs. upcoming doses.

UI Reasoning: Prioritizes time-sensitive info so caregivers see the most critical task immediately. The timeline visualizes the day's progress to reduce mental effort.



Wireframe 3: Missed Dose Alert

Design Decision: A focused Center-Aligned Bottom Sheet featuring a high-contrast "Call Mom" button as the primary action.

UI Reasoning: Reduces panic during high-stress alerts by providing an immediate communication bridge to the patient while blurring background distractions.

Wireframe 2: Daily Routine

Design Decision: Use of Differential Highlighting (red/green status tags) to compare old schedules with new pharmacist updates.

UI Reasoning: Minimizes "cognitive load" by showing exactly what changed, preventing errors during the handoff from the pharmacist.

Wireframe 4: Device Connection

Design Decision: An Illustrated Step-by-Step Guide with a central image of the hardware in "Pairing Mode".

UI Reasoning: Lowers the barrier for non-tech-savvy users by matching the app's visuals to the physical device, creating a clear sense of progress.

Section 06 - Evaluation

Evaluation Method – Feedback Collection

To validate MedTray's smart medication dispenser and understand elderly patients' medication management challenges, we collected survey responses from 40 adults at a hospital.
Participants provided feedback on:

- Daily medication routines
- Challenges in remembering doses
- Difficulties identifying pills
- Preferences for reminders and alerts
- Features that would make medication management easier and safer

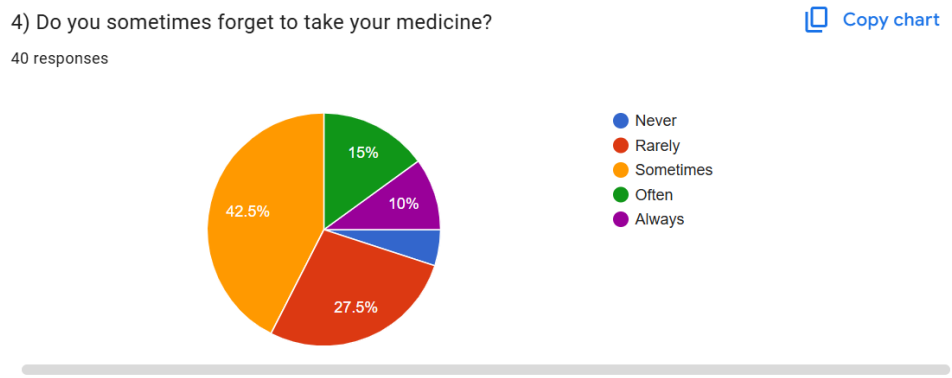
The feedback focused on usability, clarity, and accessibility for elderly users.

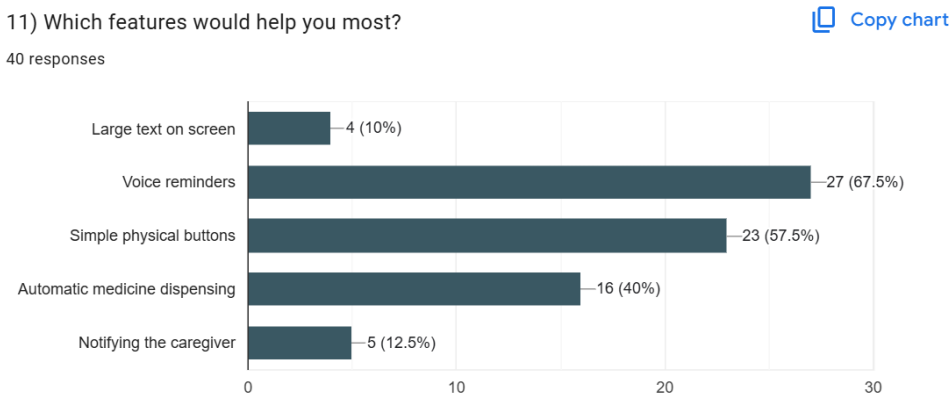
Key Findings and Design Improvements

Finding	User Feedback	The Solution	UX Principle
Forgetting doses	Sometimes forget to take medicine	Real-time reminders; caregivers are notified of missed doses	Memory support
Taking wrong medicine	Confused which pill or dose to take	Dispenser gives correct pill and dose automatically	Error prevention
Confusion with schedules	Hard to track multiple medicines	Caregiver's timeline shows updates about patient's medicine	Progressive disclosure
Difficulty reading	Small labels are hard to read	Large text display, clear labels, voice reminders	Accessibility
Anxiety about mistakes	Worry about taking wrong medicine	Step-by-step guidance; clear alerts	Memory support & error prevention

Short Reflection

The testing and survey results confirmed that elderly patients struggle with memory, pill identification, and timing, while caregivers need remote reassurance and simple controls. MedTray's integrated solution—combining automatic dispensing, visual and audio cues, and caregiver monitoring—successfully reduces errors, lowers cognitive load, and improves confidence for both patients and caregivers. Future iterations may include voice-guided instructions and customizable alerts based on user preference.





References / Links

- **Survey Link:** https://forms.gle/sUjeJBG85bGXUnfA9
- **Prototype Link:** https://med-magic-helper.lovable.app